# ABEL DOÑATE MUÑOZ

**Mathematics and Engineering Physics**

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abeldonate

## EDUCATION

### Master of Mathematics
**University of Bonn**
2025 – Bonn, Germany

### Student in CFIS program
**Polytechnic University of Catalonia (UPC)**
2020 – 2025 Barcelona, Spain

CFIS is a special excellence school created by UPC university to host some of the best mathematicians, physics and engineering students in Spain, offering them the possibility of making a double degree.

### Degree on Mathematics
**Polytechnic University of Catalonia (UPC)**
2020 – 2024 Barcelona, Spain

### Degree on Engineering Physics
**Polytechnic University of Catalonia (UPC)**
2020 – 2024 Barcelona, Spain

### Degree on Computer Engineering
**UNED**
2025 – Online

## EXPERIENCE

### Lean seminar
**Graduate seminars**
2026 (scheduled) University of Bonn

Talk about type class inference in Lean on the Lean seminar organized by the formalized mathematics department formalized mathematics department).

### Merged PRs to Mathlib
2026 Remote

Several PRs merged in Mathlib during 2026 in the area of Algebraic Topology and Commutative Algebra

### Computer Asisted Proofs Tutor
**Winter semester**
2026 University of Bonn

Tutor of the subject of computer assisted proofs (Lean).

## FURTHER EXPERIENCE

### Lean and type theory talks in MBL
**MBL math camp**
2026 Poland

Series of talks about Type Theory and Lean for high school students in the context of MBL math camp.

### Work in Mathematical Society of Catalonia
**Barcelona, Spain**
August 2025 - Remote

Work in SCM as a problem setter in some math competitions and carrying the web. I am actively part of the youth section.

### Research stay in ICMAT
**Institute of Mathematical Sciences**
2023, July Autonomous University of Madrid

Research on Bounded cohomology of groups and quasimorphisms during a stay in ICMAT (Instituto de Ciencias Matemáticas) with a JAE-INTRO scholarship. (lecture here).

### Research stay in CRM
**Mathematical Research Centre**
2022, July University of Barcelona

Research on Coxeter Matroids during a stay in CRM (Centre de Recerca Matemàtica) during Barcelona Introduction to Mathematical Research (lecture here).

### Fellow intern of the subject of computer science
**Polytechnic University of Catalonia**
2022 – 2023 Barcelona

Work as fellow intern in the subject of computer science in the degree of mathematics

### Volunteering for building software in Sierra Leone
**Lunsar, Sierra Leone**
January 2024 Saint John of God hospital

Project organized by a cooperation association inside Polytechnic University of Catalonia. Building software for a hospital in Sierra Leone.

## PROGRAMMING LANGUAGES

- Lean
- Python, C++/C#, LaTeX
- Macaulay2, Sage

# SCHOLARSHIPS

### Full Degree Scholarship

**Polytechnic University of Catalonia (UPC)**

2020 – Today  Barcelona

Full scholarship granted by the Interdisciplinary Higher Education Centre (CFIS) for the whole degree of Mathematics and Engineering Physics.

### Full Residence Scholarship

**Polytechnic University of Catalonia (UPC)**

2020 – Today  Barcelona

Full residence scholarship granted by the Interdisciplinary Higher Education Centre (CFIS) for the whole degree.

# AWARDS

**Gold Medal in Spain's Physics Olympiad**
National competition held annually in Spain for pre-university students

**Silver Medal in Ibero-American Physics Olympiad**
International competition held annually for pre-university students

**Classification to Spain's Math Olympiad**
National competition held annually in Spain for pre-university students

# MISC

**Graphic Design**
- Inkscape, Gimp, 3D-Design

**Music**
- Professional studies in piano and trumpet

# LANGUAGES

| | |
|---|---|
| English | C1 |
| Spanish | Native |
| Catalan | Advanced |
| German | Beginner |

# HOBBIES

- Dancing Salsa, Music and jamming sessions
- Sports (running, swimming, hiking, tennis, ...)
- Reading essays (history, philosophy, music, ...)
- Drawing in charcoal

# REFEREES

**Prof. van Doorn, Floris**
vdoorn@math.uni-bonn.de
University of Bonn
Professor of formalization.

**Prof. Rué, Juan José**
juan.jose.rue@upc.edu
Polytechnic University of Catalonia (UPC)
Director of Interdisciplinary Higher Education Centre (CFIS).

**Prof. Àlvarez, Josep**
josep.alvarez@upc.edu
Polytechnic University of Catalonia (UPC)
Professor of commutative algebra.

**Prof. Barja, Miguel Ángel**
miguel.angel.barja@upc.edu
Polytechnic University of Catalonia (UPC)
Professor of algebraic geometry and former director of Inderdisciplinary Higher Education Centre (CFIS).

# CONFERENCES

### Formalization and Proof Assistants

**UniDistance**

March 2026  Brig, switzerland

### ItaLean

**University of Bologna**

December 2025  Bologna, Italy

### Southwest Local Algebra Meeting

**Arizona State University**

March 2025  Phoenix, Arizona

### Hypatia graduate school

**Mathematical Research Centre (CRM)**

June 2024  Aut. Univ. of Barcelona

### Jae School

**Institute of Mathematical Sciences (ICMAT)**

July 2023  Aut. Univ. of Madrid

### Conferences at CRM

**Mathematical Research Centre (CRM)**

July 2022  University of Barcelona